Homework 4
of
STAT 5353 Probability and Statistics for Data Science and Bioinformatics

Due: 11:59 pm

March 10 (Tuesday), 2026

Problem 1 (40 points)

Continue to work on the Sleep Health and Lifestyle Dataset provided in Homework 2. Use the package `ggplot2` to visualize the data. You need to include the resulting plots and related code to get the full points. For each plot:

- make it complete/readable, in other words, it should include axis label(s), title, and legend if necessary;
- write 1–2 sentence about what the chart tells you about the data.

(a) Choose an appropriate number of bins to create a histogram of `Heart Rate`

(b) According to the data set, "Normal Weight" appears to be a typo, which would be reclassified as "Normal." Update the `BMI category` and make a scatter plot of `Sleep Duration` against `Age` and set the color to `BMI Category`.

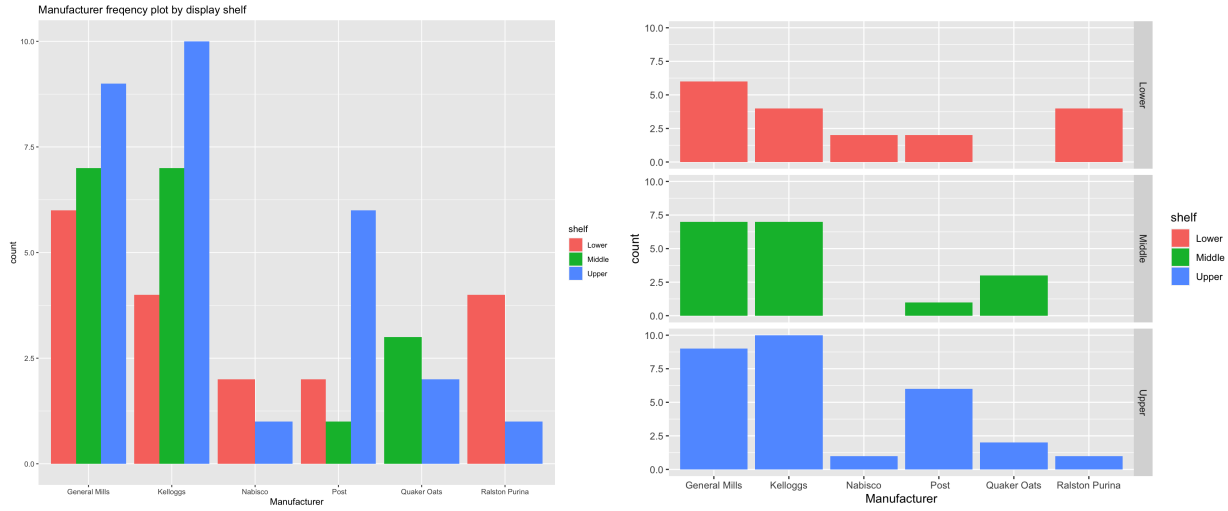
(c) Recreate the plot from part (b), and add the trend lines for each BMI Category separately.

(d) Show the relationship between `Sleep Duration` and `Sleep Disorder` by making a point, jitter, box plot, and violin plot, respectively.

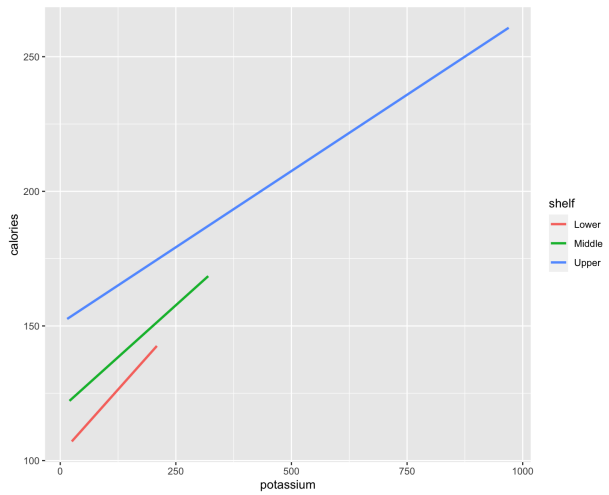
Problem 2 (30 points)

Let's work on the data set `UScereal` in the package `UsingR`. Replace the levels of the factor variable `mfr` to their full names, i.e. "G" for General Mills, "K" for Kelloggs, "N" for Nabisco, "P" for Post, "Q" for Quaker Oats, and "R" for Ralston Purina. (Hint: use the function levels()) and turn the variable shelf to a factor variable, of which levels are "1" for low, "2" for middle, and "3" for upper. Use the package `ggplot2` to visualize the data.

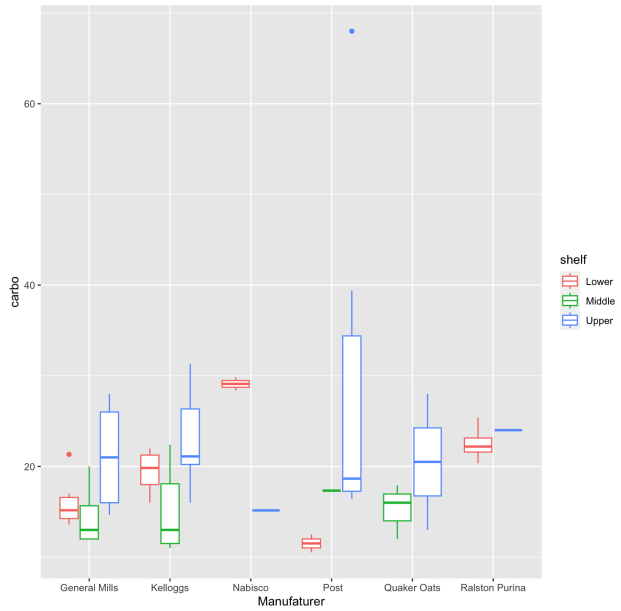
(a) Recreate the following two plots, add a short title, and comment on the merits of each one compared to the other



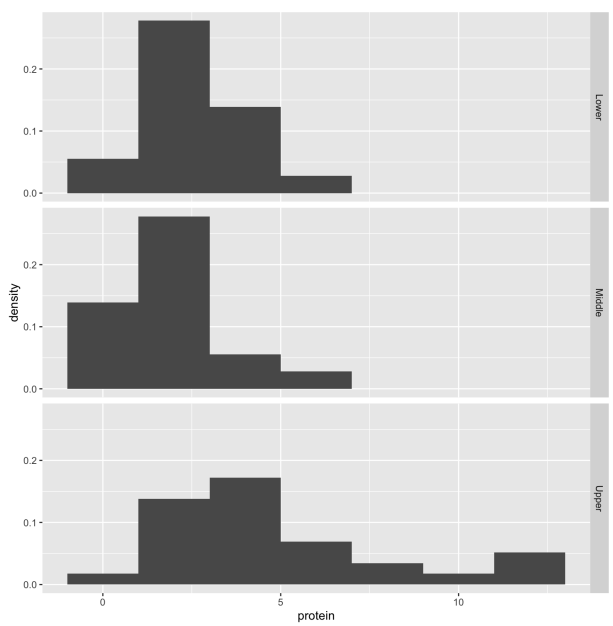
(b) Recreate the following plot and add a short title.



(c) Recreate the following plot and add a short title.



(d) Recreate the following plot and add a short title.



Problem 3 (30 points)

Download the Red Wine Quality (https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009) from Kaggle. Read the data in its original format (.csv). The description of the $p = 12$ variables are listed below.

- `fixed acidity`: Most acids involved with wine or fixed or nonvolatile (do not evaporate readily)
- `volatile acidity`: The amount of acetic acid in wine, which at too high of levels can lead to an unpleasant, vinegar taste
- `citric acid`: Found in small quantities, citric acid can add 'freshness' and flavor to wines
- `residual sugar`: The amount of sugar remaining after fermentation stops, it's rare to find wines with less than 1 gram/liter and wines with greater than 45 grams/liter are considered sweet
- `chlorides`: The amount of salt in the wine
- `free sulfur dioxide`: The free form of SO_2 exists in equilibrium between molecular SO_2 (as a dissolved gas) and bisulfite ion; it prevents microbial growth and the oxidation of wine
- `total sulfur dioxide`: Amount of free and bound forms of SO_2; in low concentrations, SO_2 is mostly undetectable in wine, but at free SO_2 concentrations over 50 ppm, SO_2 becomes evident in the nose and taste of wine
- `density`: The density of water is close to that of water depending on the percent alcohol and sugar content
- `pH`: Describes how acidic or basic a wine is on a scale from 0 (very acidic) to 14 (very basic); most wines are between 3-4 on the pH scale
- `sulphates`: A wine additive which can contribute to sulfur dioxide gas (SO_2) levels, wich acts as an antimicrobial and antioxidant
- `alcohol`: The percent alcohol content of the wine
- `quality`: Between 0 and 10

(a) Make a scatter plot of `alcohol` against `pH` and set the color to `Quality`.

(b) Make a scatter plot of `alcohol` against `pH` and facet it by `Quality`.

(c) Make a boxplot representing the relationship between the quality of wine, the volatile acidity, and different buckets of alcohol percentage $((8\% - 9\%], (9\% - 10.5\%], (10.5\% - 12\%], (12\% - 16\%])$.